

2(a)	$f_0 = f_s v / (v - v_s)$ $9560 = f \times 1510 / (1510 - 4.50)$	C1
	$f = 9530 \text{ Hz}$	A1
2(b)(i)	$v^2 = u^2 + 2as$ height = $5.6^2 / (2 \times 9.81)$	C1
	= 1.6 m	A1
2(b)(ii)	downward sloping straight line starting from a point on the speed axis and ending at point (T, 0)	B1

Question	Answer	Marks
2(b)(iii)	$(\Delta)E = mg(\Delta)h$ $= 0.45 \times 9.81 \times 1.6$	C1
	$= 7.1 \text{ J}$	A1
2(b)(iv)	air resistance increases (and weight constant)	B1
	(resultant force decreases so) acceleration decreases	B1

Question	Answer	Marks
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